

Nested List

1. List

Definition

➔ A **list** is a collection of items stored in a single variable. It is **ordered** and **changeable (mutable)**.

Syntax

```
list_name = [item1, item2, item3]
```

Example

```
fruits = ["apple", "banana", "mango"]
```

```
print(fruits)
```

Output:

```
['apple', 'banana', 'mango']
```

Features of List

- Ordered
- Changeable
- Allows duplicate values
- Different data types can be stored

Example

```
data = [10, "python", 3.5, True]
```

```
print(data)
```

Accessing Elements

```
fruits = ["apple", "banana", "mango"]
```

```
print(fruits[0])
```

```
print(fruits[1])
```

Output:

apple

banana

Modifying List

```
fruits = ["apple", "banana", "mango"]
```

```
fruits[1] = "orange"
```

```
print(fruits)
```

Output:

['apple', 'orange', 'mango']

Important List Methods

Method	Description
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append()	Adds an element
insert()	Inserts an element
remove()	Removes an element
pop()	Deletes an element
sort()	Sorts the list
reverse()	Reverses the list

Example:

```
numbers = [10, 20]
numbers.append(30)
print(numbers)
```

Output:

```
[10, 20, 30]
```

2. Nested List**Definition**

A **nested list** is a list inside another list.

Syntax

```
list_name = [ [ ], [ ], [ ] ]
```

Example

```
marks = [
    [90, 80],
    [70, 60],
    [50, 40]
]
```

```
print(marks)
```

Output:

```
[[90, 80], [70, 60], [50, 40]]
```

Accessing Nested List Elements

```
marks = [[90, 80], [70, 60], [50, 40]]
```

```
print(marks[0][1])
```

Output:

80

Explanation:

- marks[0] → [90, 80]
 - marks[0][1] → 80
-

Example

```
students = [  
    ["Ram", 21],  
    ["Ravi", 22],  
    ["Anu", 20]  
]
```

```
print(students[1][0])
```

Output:

Ravi

3. Tuple

Definition

A **tuple** is an ordered collection of items that **cannot be changed (immutable)**.

Syntax

```
tuple_name = (item1, item2, item3)
```

Example

```
colors = ("red", "green", "blue")  
print(colors)
```

Features of Tuple

- Ordered
 - Not Changeable
 - Allows duplicates
-

Accessing Tuple Elements

```
colors = ("red", "green", "blue")  
print(colors[1])
```

Output:

green

Example

```
numbers = (10, 20, 30)  
print(numbers[0])
```

Output:

10

4. Set

Definition

A **set** is an unordered collection of unique elements.

Syntax

```
set_name = {item1, item2, item3}
```

Example

```
numbers = {10, 20, 30}
```

```
print(numbers)
```

Features of Set

- Unordered
 - No duplicates
 - Changeable
 - Indexing not allowed
-

Example

```
numbers = {10, 20, 20, 30}
```

```
print(numbers)
```

Output:

```
{10, 20, 30}
```

Duplicate 20 is removed automatically.